Bhartiya Vidya Bhavan's

Sardar Patel Institute of Technology

Department of Computer Engineering

Academic Year 2026-27

Mini Project - Weekly Progress Report

Mini Project Title: Dynamic Timetable: A Role-Based Timetable Monitoring System with Automated Timetable Extraction and Natural-Language Queries

Name of Students:

1. Shivam Mulani (UID 2024300141)

2. Gurshaan Nandrajog (UID 2024300146)

3. Harshvarddhan Waghela (UID 2024300272)

Name of the Supervisor: Prof. Anant Nimkar

Review Date	Details of Work Done and Review of Meeting	Students Signature	Remarks by Supervisor	Supervisor Signature
17-08-2026	**Week 1 (Phase I evaluation): Problem identification and topic finalisation** • Studied the department's 10-page timetable PDF and listed the problems students and faculty face: manual lookup, stacked lab batches, page-wise legends, clashes found late, old and revised versions circulating together. • Presented the problem, title, objectives and scope to the guide. Work split: Shivam (architecture, database, security, query engine), Gurshaan (extraction, clash detection, performance), Harshvarddhan (mobile/web app, testing, documentation).			

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24-08-2026	**Week 2: Study of existing systems** • Studied timetable generators (FET, UniTime, aSc), PDF table extractors (Tabula, Camelot), calendar apps, ERP modules and LLM text-to-SQL; noted what each can and cannot do for a published timetable. • Collected real questions students and teachers ask about the timetable (next class, free room, teacher's location).			
02-09-2026	**Week 3 (Phase II evaluation): Literature survey and research gap** • Reviewed 15 research papers: timetabling, table recognition and OCR, natural-language interfaces to databases, prompt/SQL injection, interval reasoning, RBAC and password hashing. • Presented the literature survey, comparison table and research gap to the guide.			
07-09-2026	**Week 4: Requirements (SRS)** • Wrote functional requirements for the 5 roles (Student, Professor, HOD, Principal, Admin) with acceptance criteria. • Fixed non-functional targets: at least 95% classes verified, answer in under 300 ms, privacy of personal timetables; listed assumptions and constraints.			
14-09-2026	**Week 5: Architecture and technology selection** • Designed the overall architecture: one Expo app for Android/iOS/web, FastAPI back end, background worker, PostgreSQL as database and job queue. • Selected and tried out the stack: FastAPI, PostgreSQL 16, PyMuPDF, Tesseract OCR, React Native/Expo, Argon2id/JWT.			
21-09-2026	**Week 6: Database and API design** • Shivam: database schema (31 tables, 3NF) with ER diagram, constraints and indexes; REST API design. • Drew DFD level 0 and 1, use-case, class and sequence diagrams.			

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30-09-2026	**Week 7 (Phase III evaluation): Algorithm and module design** • Gurshaan: designed the extraction pipeline, the time-row backtracking algorithm and group-sort-sweep clash detection on sample pages of the real PDF. • Shivam: designed the question-answering engine (entity resolution, clarification, parameterised templates). Harshvarddhan: app screen flow for each role. • Presented the complete design to the guide.			
05-10-2026	**Week 8: Core implementation** • Shivam: schema and migrations, login with Argon2id and rotating JWT tokens, role-based access, admin master data, query engine. • Gurshaan: extraction from PDF, OCR, Excel and Word; background worker; archives, review and clash detection. • Harshvarddhan: Expo app with role dashboards, weekly timetable, Ask screen, uploads, review and admin screens; unit and end-to-end tests.			
12-10-2026	**Week 9 (Phase IV evaluation): Real-PDF testing and demonstration** • Tuned extraction on the college's real PDF (noon-row label, legends, combined B.E. page, LLC slots): 452 of 453 classes verified (99.8%); 542 raw overlaps reduced to the 2 genuine clashes. • Seeded the real timetable with one student login per lab group; demonstrated the working system to the guide.			
19-10-2026	**Week 10: Platforms, deployment and security** • Harshvarddhan: app runs in a laptop browser with one-click start; upgraded to Expo SDK 58 for iPhone. • Shivam: free online demo on Render; fixed a password-reset security issue found in the audit. Gurshaan: load test (0 wrong answers with 32 parallel clients, 37-47 ms median).			

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26-10-2026	**Week 11: Report, poster and improvements from review** • Wrote the report in LaTeX and designed the A3 poster. • Replaced the fixed list of ~30 question types with a compositional query planner; teachers found by code, first name, surname or 'X sir'; subject typos matched; anyone can view any teacher's, batch's or room's timetable (personal timetables stay private).			
02-11-2026	**Week 12: Final deliverables and ESE preparation** • IEEE-format research paper, 29-slide presentation, scenario-based challenges; report re-formatted to the department's Overleaf template. • 179 back-end + 20 app tests and 45 end-to-end checks passing; plagiarism and AI-content reports; demo rehearsal for the ESE on 04-11-2026.			